SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

Candidate Name

Class

CHEMISTRY **9729/01**

JC2 Preliminary Examination 20 September 2018
Paper 1 Multiple Choice 1 hour
Additional Materials: Data Booklet
Optical Mark Sheet (OMS)

READ THESE INSTRUCTIONS FIRST

On the separate multiple choice OMS given, write your name, subject title and class in the spaces provided.
Shade correctly your FIN/NRIC number.

There are **30** questions in this paper. Answer **all** questions.
For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the one you consider correct and record your choice using a **soft pencil** on the separate OMS.
Each correct answer will score one mark.
A mark will not be deducted for a wrong answer.

You are advised to fill in the OMS as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Any rough working should be done in this question paper.

Answer all questions

1 One mole of sulfuric acid is used to make an aqueous solution. The solution contains H_2SO_4 molecules, H^+ ions, SO_4^{2-} ions and HSO_4^- ions.

Which statements are correct?

(1) The solution contains 6.02×10^{23} sulfur atoms.

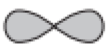
(2) The solution contains an exactly equal number of H^+ ions and HSO_4^- ions.

(3) One mole of SO_4^{2-} ions contains two moles of electrons

A	1 only
B	1 and 2 only
C	2 and 3 only
D	1 and 3 only

Answer: **A**
For statement 1 is correct: 1 mol of H_2SO_4 has 1 mol of S. The solution contains 6.02×10^{23} S atoms
For statement 2 is wrong: From the question, one mole of H_2SO_4 contains H_2SO_4 molecules, H^+, SO_4^{2-} and HSO_4^-. This means the amt of H^+ and HSO_4^- is not equal as some H_2SO_4 may have dissociate into SO_4^{2-}.
For statement 3 is wrong: One mole of SO_4^{2-} has more than 2 mol of electrons. Do not be tricked by the charge of negative two

2 The table refers to the electron distribution in the second shell of an atom with eight protons. Which row is correct for this atom?

	Orbital shape 		Orbital shape 	
	Orbital type	Number of electrons	Orbital type	Number of electrons
A	p	2	s	4
B	p	4	s	2
C	s	2	p	4
D	s	4	p	2

Answer: **B**
Second shell of an atom with eight protons: oxygen ➔ electronic configuration: $1s^2 2s^2 2p^4$

3	50 cm^3 of a 0.10 mol dm^{-3} solution of a metallic salt was found to react exactly with 25.0 cm^3 of 0.10 mol dm^{-3} aqueous sodium sulfite. In this reaction, the sulfite ion is oxidised as follows: $SO_3^{2-}(aq) + H_2O(l) \rightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2e$ What is the new oxidation number of the metal in the salt if its original oxidation number was +3?	
	A	+1
	B	+2
	C	+4
	D	+5

Answer: **B**

Amount of sulphite ions = $\frac{25}{1000} \times 0.10$ = 0.0025 mol

Amount of metallic salt = $\frac{50}{1000} \times 0.10$ = 0.005 mol

Let x be the new oxidation no of metal in salt.

[R]: $M^{3+} + (3-x)e \rightarrow M^x$

Since moles of electrons gained = moles of electrons lost in a redox reaction,

$$\frac{3-x}{2} = \frac{0.0025}{0.005}$$

x = +2

4	A 2 g sample of hydrogen at temperature T and of volume V exerts a pressure *p*. Deuterium, ${}^{2}_{1}H$ is an isotope of hydrogen. Which of the following would also exert a pressure of *p* at the same temperature T?	
	A	A mixture of 2 g of hydrogen and 2 g of deuterium of total volume 2V
	B	A mixture of 1 g of hydrogen and 2 g of deuterium of total volume 2V
	C	A mixture of 1 g of hydrogen and 2 g of deuterium of total volume V
	D	A mixture of 1 g of hydrogen and 1 g of deuterium of total volume V

Answer: **C**

P (of sample of 2 g of hydrogen) V = nRT

P (of sample of 2 g of hydrogen) $= \dfrac{nRT}{V}$

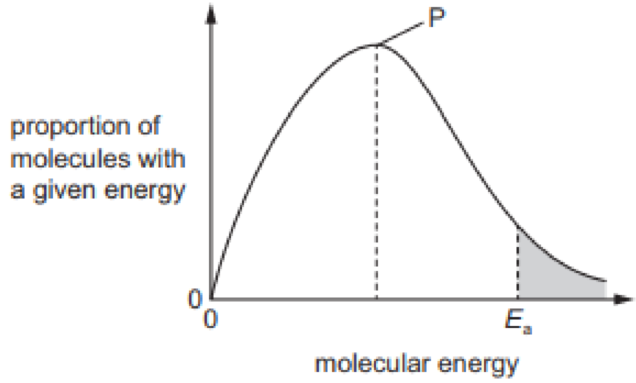
$= \dfrac{\frac{2}{2}RT}{V}$

$= \dfrac{RT}{V}$

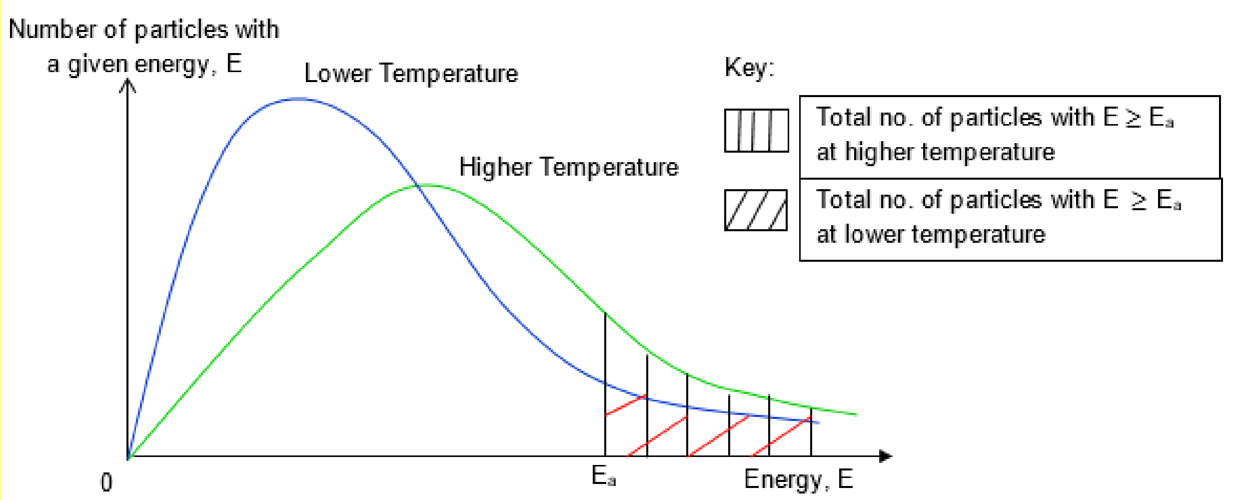
For option C:

P (of mixture) $= \dfrac{\frac{1}{2}RT + \frac{2}{4}RT}{V}$

$= \dfrac{RT}{V}$ (same pressure as the sample of 2 g of hydrogen)

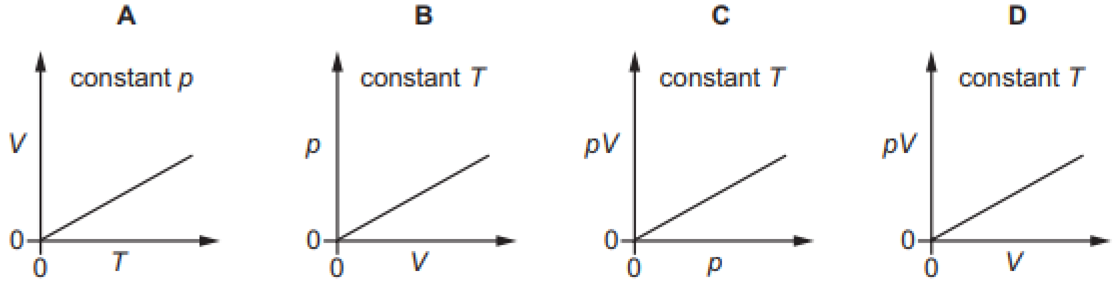
5	The diagram shows the Boltzmann distribution of energies in a gas. The gas can take part in a reaction with an activation energy, E_a.  Which statement is correct?	
	A	If temperature is increased, peak P will be lower and E_a will move to the right.
	B	If temperature is increased, peak P will be higher and E_a will not move
	C	If temperature is decreased, peak P will be the same and E_a will move to the left.
	D	If temperature is decreased, peak P will be higher and E_a will not move.

Answer: **D**



Option A and C are wrong as E_a will not shift when temperature is increased or decreased

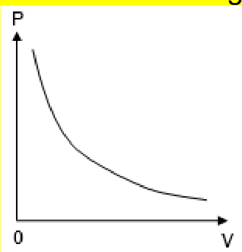
Option B is wrong. When temperature is increased, peak P will shift lower not higher.

6	Which diagram correctly describes the behaviour of a fixed mass of an ideal gas? (T is measured in K.) 

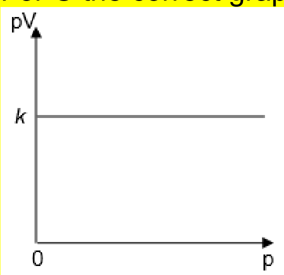
Answer: A

Using pV = nRT

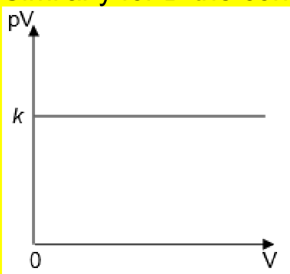
For B the correct graph is as shown



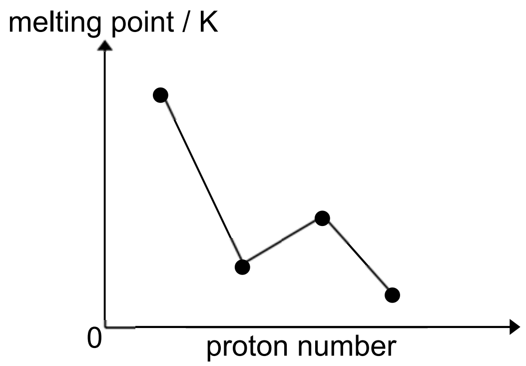
For C the correct graph is as shown



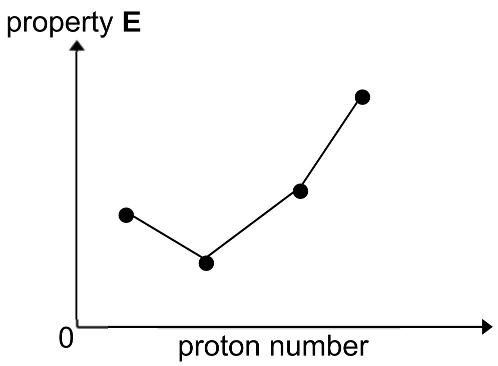
Similarly for D the correct graph is as shown

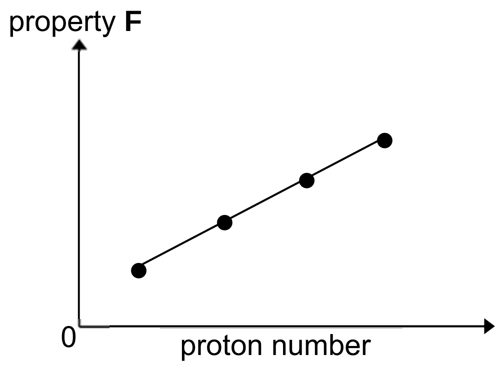


7	The diagram represents the melting points of four consecutive elements in the third period of the Periodic Table.



The sketches below represent another two properties of the elements.





What are properties **E** and **F**?

	property **E**	property **F**
A	third ionisation energy	electronegativity
B	number of valence electrons	boiling point
C	ionic radius	nuclear charge
D	electrical conductivity	atomic radius

Answer: **A**
From the melting point data, student should be able to appreciate the highest m.p should be Si and since they are consecutive elements in third period, the next 3 points will be P, S and P.

Thus, only option A is valid.

8	When 60 cm^3 of 0.1 mol dm^{-3} of sulfuric acid and 40 cm^3 of 0.2 mol dm^{-3} sodium hydroxide were mixed in a styrofoam cup, the temperature rose by 6.5 ^oC. Calculate the standard enthalpy change of neutralisation. Assume that the specific heat capacity of the solution is 4.2 $Jg^{-1}K^{-1}$.	
	A	+ 34.1 kJ mol^{-1}
	B	+ 45.5 kJ mol^{-1}
	C	- 34.1 kJ mol^{-1}
	D	- 45.5 kJ mol^{-1}

Answer: C

$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

Amount of sulfuric acid = $\frac{60}{1000} \times 0.1 = 0.006$ mol

Amount of sodium hydroxide = $\frac{40}{1000} \times 0.2 = 0.008$ mol (limiting reactant)

Amount of sodium hydroxide = amount of water = 0.008 mol

$\Delta H_{neutralisation} = -\frac{(100)(4.2)(6.5)}{0.008} = -34.1$ kJ mol^{-1}

9	In which reactions does NH_3 behave as a Brønsted-Lowry acid? (1) $HSO_4^- + NH_3 \rightarrow SO_4^{2-} + NH_4^+$ (2) $Ag^+ + 2NH_3 \rightarrow [Ag(NH_3)_2]$ (3) $2NH_3 \rightarrow NH_2^- + NH_4^+$	
	A	1 and 2 only
	B	1 and 3 only
	C	1 only
	D	3 only

Answer: D

For reaction 1, ammonia is functioning as a Brønsted-Lowry base as it received a proton from HSO_4^-

For reaction 2, ammonia is functioning as a Lewis base because it can share its lone pair of electrons with Ag^+.

For reaction 3, ammonia is functioning as both Brønsted-Lowry acid as well as Brønsted-Lowry base.

10	A current of 0.2 ampere passing for 5 hours through a solution of gold ions deposits a mass of 2.45 g of gold on the cathode. Which of these expressions gives the charge on a gold ion?	
	A	$\frac{2.45 \times 0.2 \times 5 \times 60 \times 60}{197 \times 96500}$
	B	$\frac{0.2 \times 5 \times 60 \times 60 \times 197}{96500 \times 2.45}$
	C	$\frac{2.45 \times 96500}{197 \times 0.2 \times 5 \times 60 \times 60}$
	D	$\frac{197 \times 0.2 \times 5 \times 60 \times 96500}{2.45}$

Answer: B

Q = I x t
= 0.2 x 5 x 60 x 60

$$\text{Amt} = \frac{I \times t}{nF}$$

$$n = \frac{I \times t}{\text{amt} \times F} = \frac{0.2 \times 5 \times 60 \times 60}{\frac{2.45}{197} \times 96500} = \frac{0.2 \times 5 \times 60 \times 60 \times 197}{2.45 \times 96500}$$

11	Pure nitrosyl chloride, NOC*l* gas, was heated at 320°C in a 2.0 dm^3 vessel. At equilibrium, 30% of the NOC*l* gas had dissociated according to the equation below and the total pressure was P atm. $2NOCl\ (g) \rightleftharpoons 2NO\ (g) + Cl_2\ (g)$ What is value of the equilibrium constant, K_p?

A	$\frac{17.9}{p}$	**B**	$\frac{41.7}{p}$	**C**	0.0120p	**D**	0.0130p

Answer: **C**

	$2NOCl\ (g)$	$\rightleftharpoons 2NO\ (g)$	$+ Cl_2\ (g)$
Initial partial pressure/atm	**x**	0	**0**
Change in partial pressure / atm	-0.3x	+0.3x	+0.15x
Equilibrium partial pressure / atm	0.7x	0.3x	0.15x

0.7x + 0.3x + 0.15x = p
x = 0.8696p

$K_p = \frac{(0.2609p)(0.1304p)^2}{(0.6087p)^2}$
= 0.01197
= 0.0120p

12	Calculate the resultant pH of the solution when 10 cm^3 of hydrochloric acid with a concentration of 0.015 mol dm^{-3} was added to a 25 cm^3 sample of ammonia with a concentration of 0.25 mol dm^{-3}. (K_b of ammonia = 1.778 x 10^{-5} mol dm^{-3})	
	A	10.9
	B	8.25
	C	7.64
	D	9.25

Answer: **A**

$HCl + NH_3 \rightarrow NH_4Cl$

Amt HCl given = 10/1000 x 0.015 = 0.00015 mol
Amt NH_3 given = 25/1000 x 0.25 = 0.00625 mol

All the HCl added will be neutralised by the excess NH_3 forming the NH_4^+ thus

[HCl ≡ NH_4^+]

Amt NH_4^+ present = 0.00015 mol

Amt of NH_3 remaining = 0.00625 – 0.00015
= 0.0061 mol

Thus, present of a basic buffer

$pOH = pK_b + \lg([NH_4^+] / [NH_3])$
$= -\lg(1.778 \times 10^{-5}) + \lg$
$= 3.14$
$pH = 14 – 3.14 = 10.9$

13	Hydrogen can be made from steam according to the following equation:

$$H_2O(g) + C(s) \rightarrow H_2(g) + CO(g)$$

The Gibbs free energy change of reaction at two different temperature are shown

ΔG_1 = +78 kJ mol^{-1} at 378 K
ΔG_2 = −58 kJ mol^{-1} at 1300 K

Which row of the table gives the correct sign of ΔH and ΔS for this reaction?

	ΔH	ΔS
A	−	−
B	−	+
C	+	−
D	+	+

Answer: D

Δn of gas = 2-1 = +1
ΔS is positive.
$\Delta G = \Delta H - T\Delta S$
As temperature increases to 1300 K, ΔG is negative and reaction is spontaneous.

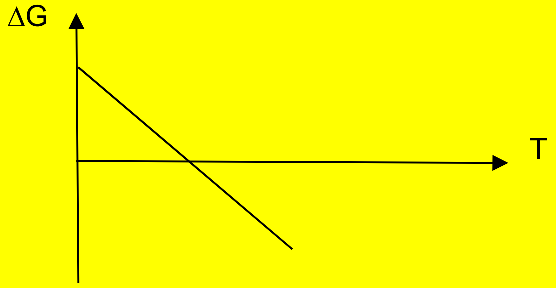
When the temperature is lower at 378 K, ΔG is positive and reaction is non-spontaneous.

Since ΔS is positive, this indicate that ΔH is positive as only low temperature can allow ΔG to become positive.

<u>Alternatively</u>, using $\Delta G = \Delta H - T\Delta S$ where

- ΔG (y-axis)
- ΔH (y intercept)
- T (x-axis)
- - ΔS (gradient)

Since ΔS is positive (which leads to a negative gradient) and ΔG changes from positive to negative with increasing temperature, ΔH is positive.



14 An experiment was carried out to investigate the initial rate of reaction between potassium peroxodisulphate, $K_2S_2O_8$, an oxidising agent, and potassium iodide, KI.

The initial volumes of the $K_2S_2O_8$ and KI solutions in the mixture together with the time taken for the mixture to darken for the various experimental runs are given below.

Volume of $K_2S_2O_8$ / cm^3	Volume KI / cm^3	Volume of water / cm^3	time taken to darken / s
10	20	10	35
5	20	15	70
10	8	22	88
20	40	20	***y***

Select the correct option for the following reaction.

	Order with respect to $K_2S_2O_8$	Order with respect to KI	**y** / s
A	1	1	70.0
B	1	2	17.5
C	1	1	35.0
D	2	1	17.5

Answer: **C**
From the first two expt, order wrt $K_2S_2O_8$ is one.
From the 1st and 3rd expt, order wrt is KI one.
In the 4th expt, the concentration of $K_2S_2O_8$ and KI are both doubled from experiment one respectively. However, since total volume is doubled too, the concentrations of expt 4 are exaxtly the same as expt one. So the time taken for the solution to darken is the same.

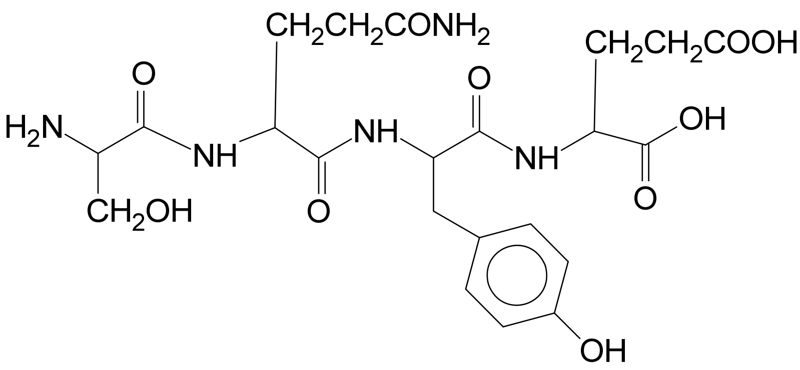
15	The enzyme maltase speeds up the reaction between maltose and water. $$\text{maltose + water} \xrightarrow{\text{maltase}} \text{glucose}$$ Maltase shows **specificity**. Which statement describes the **specificity** of maltase?	
	A	Maltase is a biological catalyst and it is a type of protein.
	B	Maltase is most effective between pH 6.1 and pH 6.8.
	C	Maltase lowers the activation energies of the reactions it catalyses.
	D	Maltase only speeds up a small number of chemical reactions.

Answer: **D**
Being a biological catalyst and a type of protein does not define the term **specificity.** Thus option A is out

Effectiveness over a pH range does not define the term **specificity**. This make option B wrong.

Lowering activation energy does not define the term **specificity**. In fact all catalyst or enzyme lower E_a. Option C is thus wrong.

16	The diagram shows the structure of the tetrapeptide, **J**.



Which statements are correct?

(1) When 1 mol of **J** reacts with hot NaOH(aq) until no further reaction occurs, 8 mol of NaOH will react.

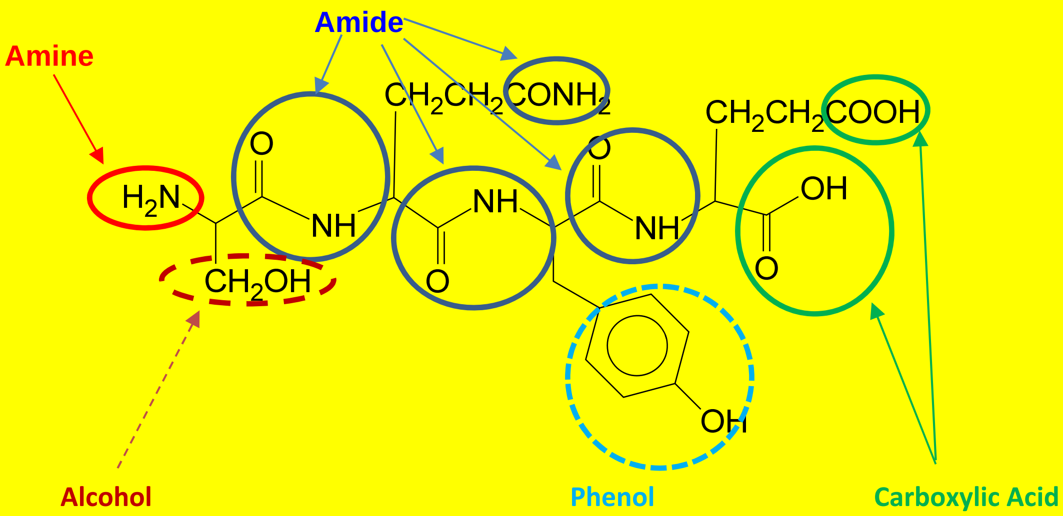
(2) When 1 mol of **J** reacts with hot HC*l*(aq) until no further reaction occurs, 5 mol of HCl will react.

(3) When 1 mol of **J** reacts with ethanoyl chloride, 3 mol of ethanoyl chloride will react forming ester or amide.

(4) When 1 mol of **J** reacts with Na(s), 4 mol of hydrogen gas will be given out.

A	1 and 2 only
B	2 and 3 only
C	1, 2 and 4 only
D	3 and 4 only

Answer: **B**

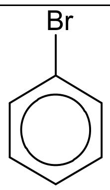
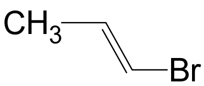
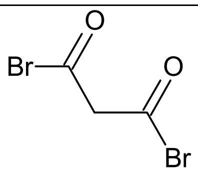


Hence, 1 mol of **J** will react with 7 mol of NaOH (aq) (Basic hydrolysis of amides, neutralisation of carboxylic acid and phenols).

1 mol of **J** will react with 5 mol of HCl (aq) (Acidic hydrolysis of amides, neutralisation of amine).

When 1 mol of **J** reacts with ethanoyl chloride, 3 mol of ethanoyl chloride will react (alcohol, phenol and amine)

When 1 mol of **J** reacts with Na(s), 2 mol of hydrogen gas will be given out. (Phenol, alcohol and carboxylic acid)

17	Which of these will produce the most silver bromide precipitate when a 1 g sample reacts with excess hot sodium hydroxide, followed by silver nitrate solution?	
	A	 (M_r: 156.9)
	B	 (M_r: 120.9)
	C	 (M_r: 229.8)
	D	CH_3Br (M_r: 94.9)

Answer: **D**

Both A and B are resistant to nucleophilic substitution due to the double bond character between C and Br.

Amt of C in 1g = $\frac{1}{229.8}$ = 0.00435 Hence amt of AgBr formed = 0.0087 mol	Amt of D in 1g = $\frac{1}{94.9}$ = 0.0105 Hence amt of AgBr formed = 0.0105

18 Identify the final product **L** in this sequence of reactions.

$$CH_2CHCOCH_2CHO \xrightarrow[\text{then } H_2O]{NaBH_4 \text{ in } CH_3OH} \mathbf{K} \xrightarrow[\text{heat}]{K_2Cr_2O_7/\ H^+} \mathbf{L}$$

A	$CH_2CHCOCH_2COOH$
B	$CH_3CH_2COCH_2COOH$
C	$CH_2(OH)CH(OH)COCH_2CH_2OH$
D	$CH_2(OH)CH(OH)CH(OH)CH_2CH_2OH$

Answer: **A**

$$CH_2{=}CHCOCH_2CH{=}O \xrightarrow[\text{then } H_2O]{NaBH_4 \text{ in } CH_3OH} CH_2{=}CHCH(OH)CH_2CH_2OH$$

$$\Big\downarrow \; K_2Cr_2O_7/\ H^+,\ \text{heat}$$

$$CH_3CH_2COCH_2COOH$$

19 An alcohol **M** with molecular formula $C_4H_{10}O$ is oxidised by acidified potassium dichromate(VI) under certain conditions to give **N**.

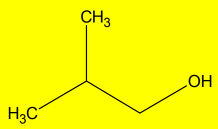
- **N** does not produce a yellow precipitate with aqueous alkaline iodine
- **N** gives a reddish brown precipitate when reacted with Fehling's solution

How many isomers of alcohol **M** could result in the observations for **N**?

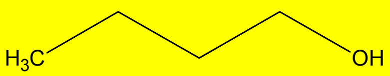
A	1
B	2
C	3
D	4

Answer: **B**

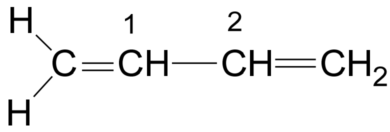
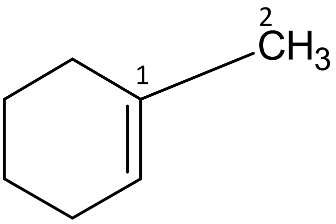
Since product **N** gives a reddish brown precipitate when reacted with Fehling's solution, an aldehyde functional group is present. Since aldehydes are formed from controlled oxidation of primary alcohol, the possible structures of primary alcohol from $C_4H_{10}O$ are:



and



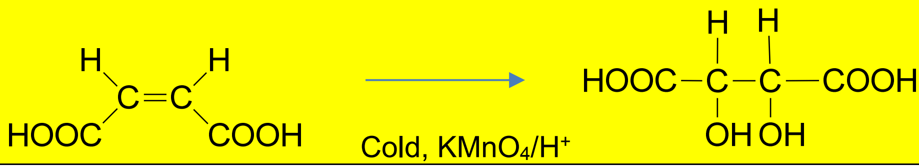
Therefore, there are 2 isomers.

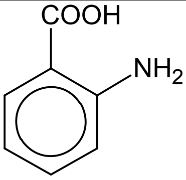
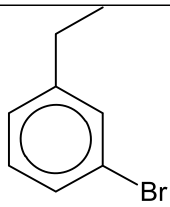
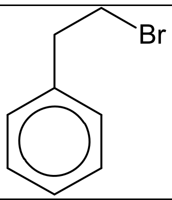
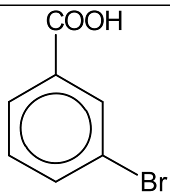
20	Which of the following compounds has the shortest C1-C2 bond length?
A	1 2 $CH_3—CH_2—CH_3$
B	1 2 $CH_3—CH=CH_2$
C	
D	

Answer: C
Percentage of s character of hybridised orbitals are in order sp > sp^2 > sp^3. The higher the percentage of s character, the shorter the bond formed as the hybridised orbital will be more spherical.
Option A has sp^3-sp^3 overlap.
Option B and D has sp^3-sp^2.
Option C has sp^2-sp^2 overlap.

21	Maleic acid is used in the food industry and for stabilising drugs. It is the cis-isomer of butenedioic acid and has the structural formula $HO_2CCH=CHCO_2H$. What is the product formed from the reaction of maleic acid with cold, dilute, acidified manganate(VII) ions?
A	$HO_2CCH(OH)CH(OH)CO_2H$
B	HO_2CCO_2H
C	$HO_2CCH_2CH(OH)CO_2H$
D	$HO_2CCOCOCO_2H$

Answer: **A**

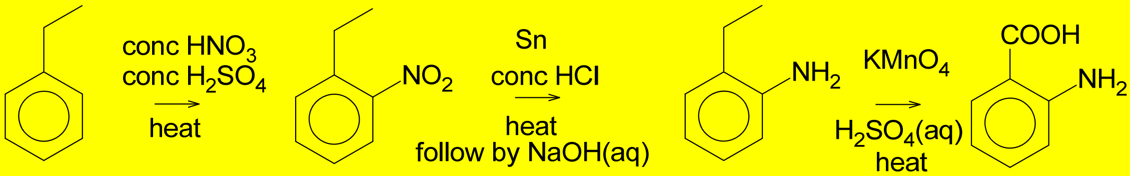


22	Which one of the following compounds **cannot** be synthesised from ethylbenzene?			
	A		**B**	
	C		**D**	

Answer: **B**

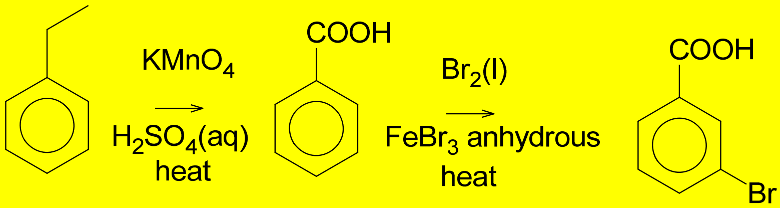
Ethyl group is 2,4 directing. Thus option B is wrong

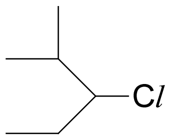
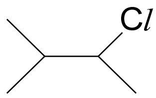
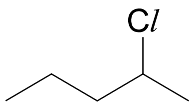
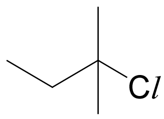
Option A is correct



Option B is correct. Side chain free radical substitution has occurred.

Option D is correct

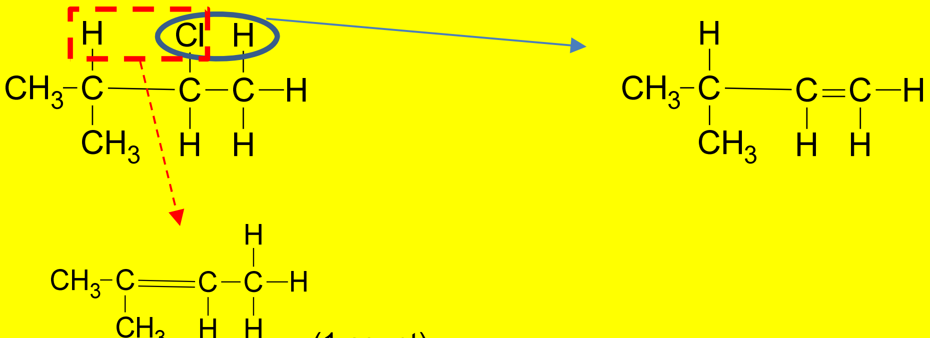


23	Structural isomerism and stereoisomerism should be considered when answering this question. A colourless liquid, $C_5H_{11}Cl$, exists as a mixture of two optical isomers. When heated with sodium hydroxide in ethanol, a mixture of only **two** alkenes is formed. What could the colourless liquid be?			
	A		**C**	
	B		**D**	

Answer: **C**

Option A is wrong as it has one more carbon and thus does not match the molecular formula. Option D is out as they do not have chiral carbon to allow presence of two optical isomers.

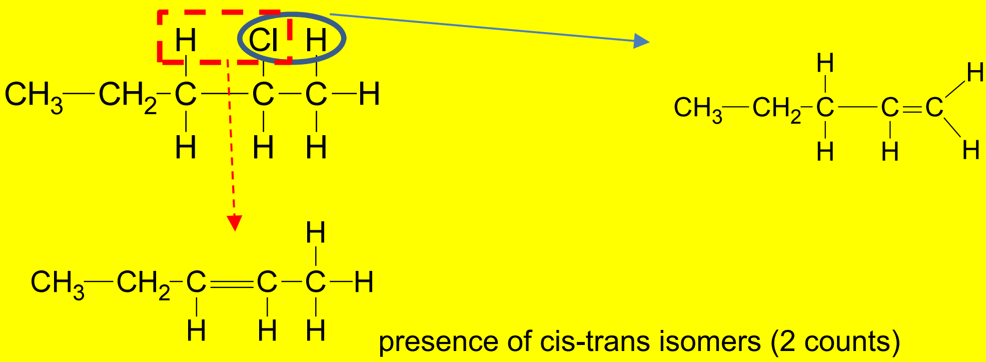
Option C is correct



(1 count)

(1 count)

Option B is wrong as during the elimination process (hot sodium hydroxide in ethanol) a mixture of three alkenes were obtained

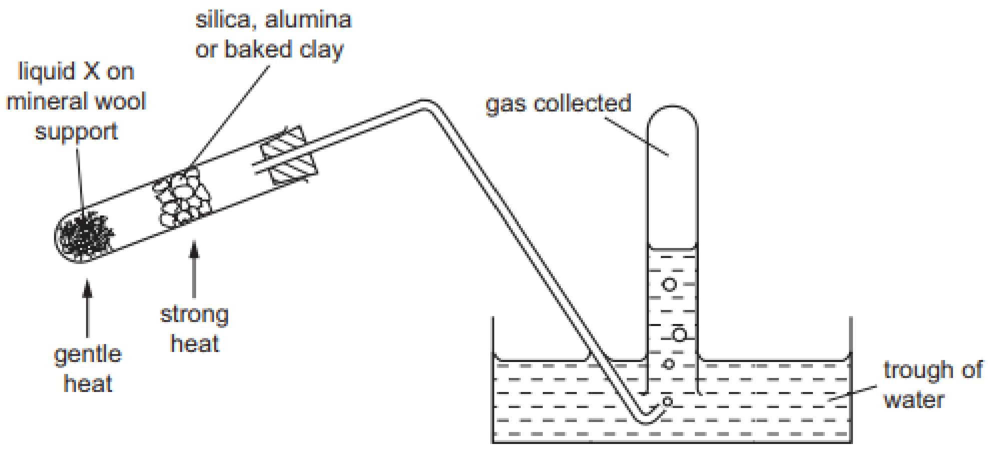


(1 count)

presence of cis-trans isomers (2 counts)

Thus, Option B structure will result in a mixture of 3 alkenes.

24	The diagram shows an experimental set-up which can be used in several different experiments.



Which processes could be demonstrated by using the above apparatus?

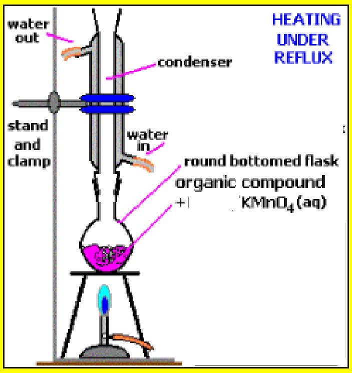
(1) oxidation of ethanol (liquid X)

(2) dehydration of ethanol (liquid X)

(3) cracking of paraffin (liquid X)

A	1, 2 and 3
B	1 and 2 only
C	2 and 3 only
D	3 only

Answer: C

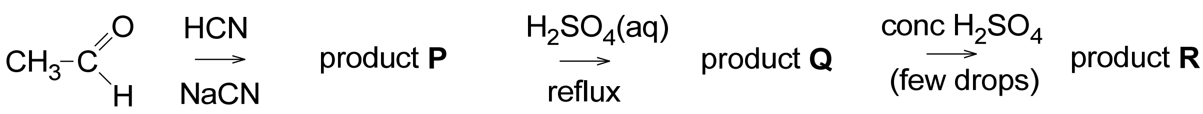


Oxidation of ethanol usually require a reflux set up

From lecture notes:

- Catalytic cracking is used to produce petrol (C_5 to C_{10}) and aromatic hydrocarbons.
- Catalyst: **zeolites** (mixture of Al_2O_3 and SiO_2).
- Temperature: **450 - 550°C** (Notice that lower temperatures are used)

25 Ethanal, CH_3CHO, is used to make product **R** in a three-stage synthesis.



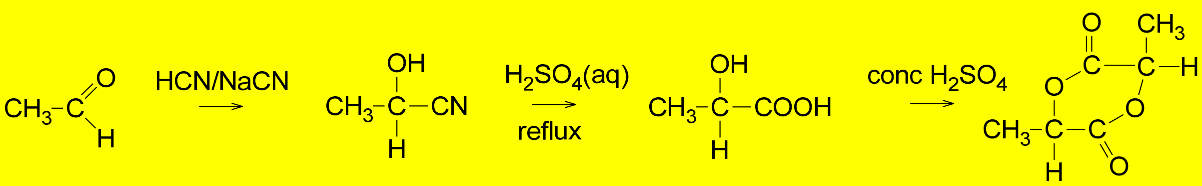
Two molecules of **Q** react to give one molecule of **R** plus two molecules of water.

R does not react with sodium.

What is the molecular formula and empirical formula of **R**?

	Molecular formula	Empirical formula
A	$C_3H_4O_2$	$C_3H_4O_2$
B	$C_6H_8O_4$	$C_3H_4O_2$
C	$C_3H_5O_2$	$C_3H_5O_2$
D	$C_6H_{10}O_5$	$C_6H_{10}O_5$

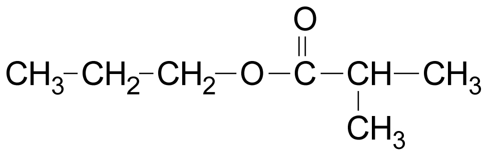
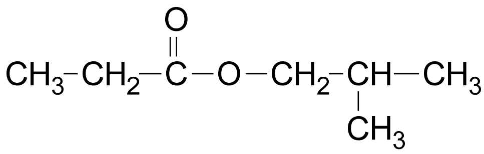
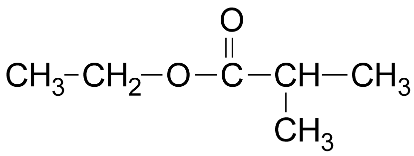
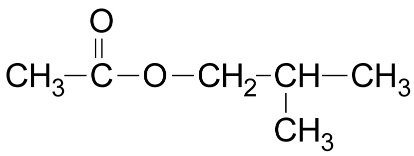
Answer: **B**



Molecular formula is $C_6H_8O_4$

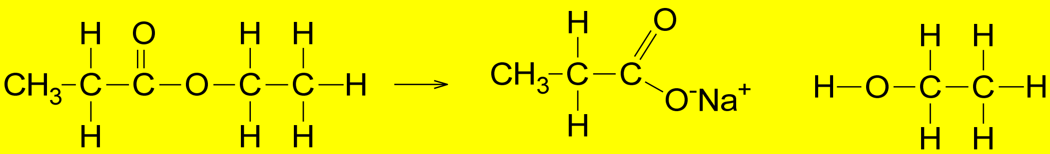
Empirical formula of product R is $C_3H_4O_2$

26	Chloroflouroalkanes have been used as the refrigerant in refrigerators but care has to be taken in disposing of old refrigerators. Which statements about chlorofluoroalkanes are correct? (1) C–C*l* bonds more readily undergo homolytic fission than C–F bonds. (2) Care is taken in the disposal of old refrigerators because of possible ozone depletion. (3) C_2H_4ClF is more volatile than C_2H_6.	
	A	2 only
	B	1 and 2 only
	C	2 and 3 only
	D	1, 2 and 3
	Answer: B C-C*l* bonds are weaker than C-F bonds thus C-C*l* bonds require less energy to break when undergoing homolytic fission. Thus statement 1 is correct. Presence of Chlorofluoroalkanes will result in ozone depletion. Statement 2 is right. Statement 3 is wrong as C_2H_4ClF is polar and there are presence of stronger intermolecular permanent dipole-permanent dipole interaction as compared to the weaker instantaneous dipole-induced dipole interaction present in the non-polar molecule of C_2H_6.	

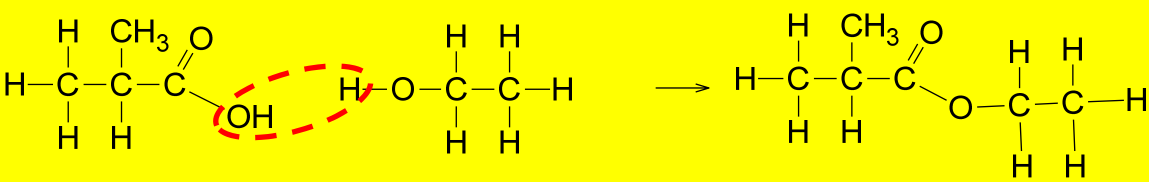
27	Ethyl propanoate is refluxed with aqueous sodium hydroxide. The alcohol produced is then reacted with methyl propanoic acid to make a second ester. What is the structural formula of this second ester?	
	A	
	B	
	C	
	D	

Answer: **C**

During the basic hydrolysis of ethyl propanoate with NaOH (aq)

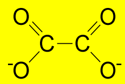


Ethanol is produced



28	Ethanedioic acid has the formula HO_2CCO_2H. What is the formula of aluminium ethanedioate?	
	A	AlC_2O_4
	B	$Al(C_2O_4)_3$
	C	$Al_2C_2O_4$
	D	$Al_2(C_2O_4)_3$

Answer: **D**



Species: Al^{3+}

Charge: +3 -2

Formula $Al_2(C_2O_4)_3$

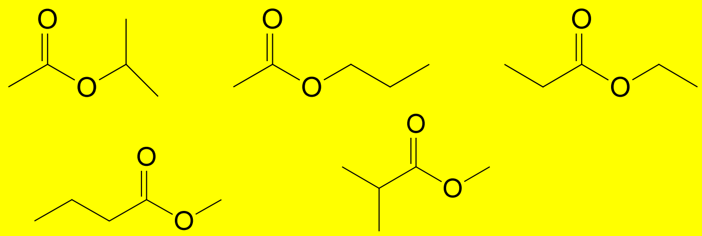
29	Which of the following processes lead to an increase in entropy? (1) Diffusion of air fresher in the lecture theatre. (2) Combustion of a piece of charcoal to form CO_2 (g) and H_2O (g). (3) Desalination of sea water by reverse osmosis (solvent passes from a dilute solution to a concentrated solution).	
	A	1 only
	B	1 and 2 only
	C	2 and 3 only
	D	1, 2 and 3

Answer: **B**

For option 3, there is an increase in orderliness as the solvent passes from a more concentrated solution to a more diluted solution. Hence, entropy will decrease.

30	How many structural isomers with the molecular formula $C_5H_{10}O_2$ give infra-red absorptions both at approximately 1300 cm^{-1} and at approximately 1740 cm^{-1}?	
	A	3
	B	5
	C	7
	D	9

Answer: **B**
Infra-red absorption of 1300 cm^{-1}: carboxylic acid and ester
Infra-red absorption of 1740 cm^{-1}: ketone, aldehyde and ester
Thus, it has to be an ester since both conditions **MUST** be met.



END OF PAPER 1